BARNES UI — the mathematics behind the analyses

Every formula used to size, simulate and choose the Type A and Type B finger joints; where each one lives in the code; a worked example with this project's own numbers; and the decision it feeds. Companion to BARNES_print_decision_report.pdf.

0. How to read this

- Each section answers one question (how stiff, how far, how much tendon force, when does it buckle, which one wins). It gives the equations, then a **worked example** with real inputs, then **where in the code** it is, then what decision it feeds.
- The order is the order you would use them on a new design: material first, then hand models (beams, tendon, buckling) to screen ideas in minutes, then FEA for the few that survive, then the selection rules.
- Units throughout: mm, N, MPa (= N/mm²), N·mm for moments and torques, radians inside formulas, degrees in tables.
- Symbols are collected in the appendix.

1. Material, strain and the E-scaling rule

1.1 The TPU model: Neo-Hookean

TPU is rubbery: it stretches tens of per cent and stays elastic, so a linear Hooke's-law model is wrong at the strains here. The simplest model that handles large stretch is Neo-Hookean. It has one stiffness constant C_{10} plus a bulk term that keeps the volume almost constant:

$$W = C_{10}\,(\bar{I}_1 - 3) + \tfrac{1}{D_1}\,(J-1)^2, \qquad \bar{I}_1 = J^{-2/3}\,\mathrm{tr}(F^T F), \quad J = \det F$$

F is the deformation gradient (how each little cube of material is stretched and turned). The constants come from the small-strain Young's modulus E and Poisson's ratio ν:

$$\mu_0 = 2C_{10} = \tfrac{E}{2(1+\nu)}, \qquad K = \tfrac{E}{3(1-2\nu)}, \qquad D_1 = \tfrac{2}{K}$$

With E = 12 MPa and ν = 0.4995 (nearly incompressible): μ_0 = 4.0027 MPa, C_{10} = 2.0013 MPa. In a uniaxial pull with stretch λ = 1 + ε the engineering stress is

$$\sigma_{eng} = 2C_{10}\,(\lambda - \lambda^{-2})$$

which is E·ε for small ε and softens gently above about 20 %. **Where in the code:** src/barnes_ui/core/material/models.py

1.2 Why every force and torque scales exactly with E

Every stress term in W is proportional to E ($C_{10} \propto E$ and $1/D_1 \propto E$ at fixed ν). Write the stress as σ = E·f(F). Equilibrium with no body force is ∇·σ = 0.

- **Displacement control** (a joint bent to an angle): the boundary fixes the motion, so F is the same field for any E (the E cancels out of ∇·(E f(F)) = 0). Strains, shapes, range of motion and centre drift do not depend on E. Every reaction (moment, torque, stiffness, stress) is E × something fixed.
- **Force control** (round-2 tendon load T): ∇·(E f(F)) = 0 with traction T on the boundary is the same problem as ∇·f(F) = 0 with traction T/E. So the deformation depends only on T/E, and a capacity found at 12 MPa is $T_{cap}(E) = T_{cap}(12) \cdot E/12$. Contact keeps this true only if the penalty stiffness is also ∝ E, which is why it is set to 10·E/h (section 5.5).

$$X(E) = X(E_{run}) \cdot \frac{E}{E_{run}} \qquad \text{for any force, torque, stiffness, stress, capacity}$$

Different studies were run at different E (A3 and Type B at 12, A6 at 13, A8 at 19.3 MPa), so they were all brought to one basis with this formula. SUNLU publishes no modulus, so the working value is 12 MPa, bracketed by 5.3 MPa (printed TPU 90A, Bambu datasheet, XY) and 19.3 MPa (bulk TPU 90A, Jabil). **Where in the code:** scripts/sunlu_rescale.py

Worked example: rescaling two headline numbers
A8 with collars, sagitta 2.75, was run at 19.3 MPa: torque at 90° = 2.809 N·mm.
At 12 MPa: 2.809 × 12 / 19.3 = **1.75 N·mm**. At 5.3 MPa: 0.77 N·mm.
B5 t2.0 deep tendon capacity at 0° was found at 12 MPa: 30.2 N. At 5.3 MPa: **13.4 N**; at 19.3 MPa: 48.6 N.
When your tensile test gives a real E, set E_SUNLU_MPA in scripts/sunlu_rescale.py and every number updates. Nothing needs to be re-simulated.

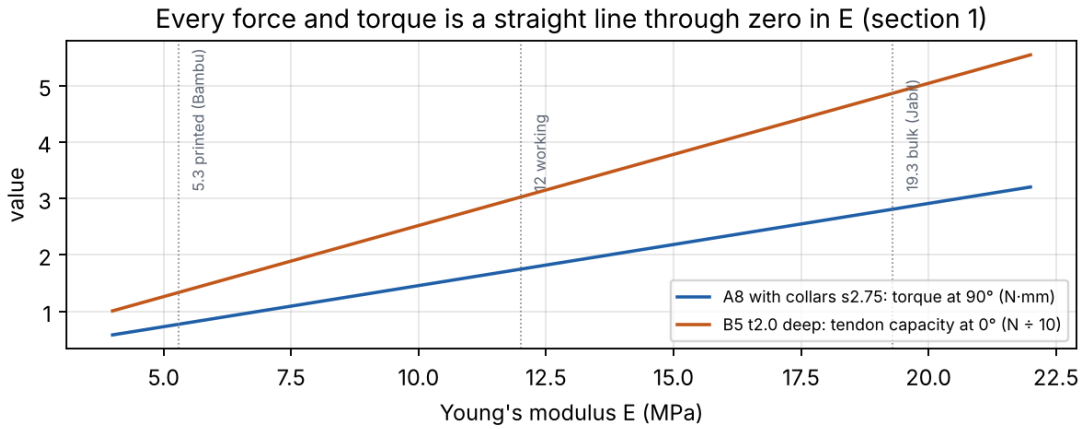


Figure 1. The E-scaling rule applied to two results. The bracket is the real uncertainty until the specimens are tested.

1.3 Which strain, and the 20 % limit

The solver (CalculiX, large-deformation mode NLGEOM) reports the Green–Lagrange strain tensor. It is turned into the largest principal engineering strain, the number a tensile test gives:

$$\mathbf{E}_{GL} = \tfrac{1}{2}\,(F^{T}F - I), \qquad \lambda_{max} = \sqrt{1 + 2\,e_{max}(\mathbf{E}_{GL})}\,, \qquad \varepsilon = \lambda_{max} - 1$$

e_{max} is the largest eigenvalue. A 0.4 mm band at each collar root is left out of the peak: the sharp inside corner there is a mathematical singularity whose value keeps rising as the mesh is refined, so it does not converge to a real number. A real print has a small radius there. That is why a fillet is recommended wherever a strip meets a collar. The 20 % limit is a placeholder for fatigue (10 % and 30 % are drawn for reference) until the printed filament has been cycled. **Where in the code:** scripts/a8_meander_study.py, TYPE B/analysis/fea_typeb.py

2. Beam models: the minutes-long screening tools

2.1 Bending strain, and why thickness is the strain knob

A strip of thickness t bent from curvature κ_0 to κ has surface strain

$$\varepsilon = \tfrac{t}{2}\,|\kappa - \kappa_0| \qquad (\text{wrapping flat around a radius } R:\ \varepsilon \approx \tfrac{t}{2R})$$

and bending stiffness EI with I = w t³/12 for a rectangle of width w. So torque grows with t³ but strain with t: doubling thickness buys 8× the torque at 2× the strain. Width scales torque linearly and leaves strain alone, as long as the strip does not start bending sideways (which is what happened to wide A8 strips). This is the main sizing rule for every spring here: **choose t from the strain limit, then w for the torque**.

2.2 The planar elastica (A6 sliding leaf)

A leaf that bends through large angles but does not stretch. Along arc length s, with tangent angle θ(s), internal moment M(s) and a constant internal force (n_y, n_z):

$$\frac{dy}{ds} = \cos\theta, \quad \frac{dz}{ds} = \sin\theta, \quad \frac{d\theta}{ds} = \kappa_0 + \frac{M}{EI}, \quad \frac{dM}{ds} = -(n_z\cos\theta - n_y\sin\theta)$$

It is solved by shooting: guess the anchor moment M_0, the force components and the free length, integrate (RK4, 400 steps), and adjust the guesses (least squares) until the far end (1, 2) lands in the guide mouth, (3) points along the guide, and (4) carries no force along the guide, since a frictionless slot cannot push along itself. Condition 4 is what makes the end slide. When the T-head hits its stop, the free length is fixed instead and condition 4 is dropped. Each angle starts from the previous solution (continuation). With EI = 1 the shape does not depend on stiffness at all, so EI is applied afterwards.

Torque comes from energy, which avoids sign mistakes:

$$U(\phi) = \int_0^L \frac{M^2}{2EI}\,ds, \qquad \tau(\phi) = \frac{dU}{d\phi}, \qquad \bar{\tau} = \frac{U(\phi_{max})}{\phi_{max}}$$

φ is the joint angle. The figure of merit for a return spring is the mean torque τ̄ over 0–105°, taken at the worst of four assumed axis positions, because the real PIP axis relative to the cuffs is not known. Pass rules: reaches 105°, restoring (τ > 0) at every angle, at least 3.5 mm from the axis, peak strain ≤ 20 %. **Where in the code:** scripts/a6_rom_check.py, scripts/a6_parameter_sweep.py

2.3 The extensible elastica (A3 twin strap) and why A3 cannot work as laid out

A3's straps are clamped at both ends and the distal clamp swings around the joint axis, so the distance between the ends changes. The arch has only arc − chord = 28.38 − 27.80 = 0.58 mm of slack. Beyond a few degrees the strap must stretch. Add axial strain N/EA to the elastica:

$$\frac{dy}{ds_0} = \Lambda\cos\theta, \quad \frac{dz}{ds_0} = \Lambda\sin\theta, \quad \Lambda = 1 + \frac{n_y\cos\theta + n_z\sin\theta}{EA}, \qquad \frac{EA}{EI} = \frac{12}{t^2}$$

For t = 1 mm, EA/EI = 12 mm⁻²: stretching is 12× stiffer than bending per unit, so a strap forced to stretch becomes very stiff and very strained. That is why the sweep of 480 A3 variants found none that passes. A sliding end (A6) or a meander (A8) removes the stretch. **Where in the code:** scripts/a3_parameter_sweep.py

2.4 Linear curved beam (Castigliano), and its check against FEA

For a quick stiffness at 0°: treat each strap as a curved beam clamped at both ends (3 unknown end actions in plane), release one end and build the flexibility matrix from unit loads:

$$C_{ij} = \int \left(\frac{m_i m_j}{EI} + \frac{n_i n_j}{EA} + \frac{k\,v_i v_j}{GA}\right) ds, \qquad \mathbf{x} = C^{-1}\mathbf{d}$$

d is the end displacement imposed by rotating the distal clamp about the joint axis; x are the end forces and moment; the restoring moment is their moment about the axis. The 3D version for meanders adds torsion (GJ) and the second bending axis, a 6×6 matrix, because a meander's runs parallel to the axis twist rather than bend.

Worked example: how sensitive a beam model is to its inputs (A3)

FEA joint stiffness of A3 1.0 mm: 6.103 N·mm/rad (mesh-converged to 0.94 %).
Beam model at the FEA's own axis and clamp line: 8.96 % to 20.2 % stiffer, depending on where exactly the strap is taken to be clamped.
Moving the assumed axis 2 mm (bounding-box centre z = 7.42 instead of the mounting features' z = 5.44) changes the beam result from 6.86 to 3.16 N·mm/rad.
Lesson: the joint axis and the clamp line matter more than the beam theory. Always state both.

Where in the code: src/barnes_ui/core/mechanics/curved_beam.py, src/barnes_ui/core/mechanics/spatial_beam.py, scripts/a3_rom_vs_fea.py

3. Tendon mechanics: how much pull a joint needs

3.1 What the servo can give

$$T_{avail} = \frac{\eta\,\tau_{stall}}{r_{horn}} = \frac{0.5 \times 0.94\ \text{N m}}{0.010\ \text{m}} = 47\ \text{N}, \qquad \text{stroke} = r_{horn}\,\phi_{servo} = 10\pi = 31.4\ \text{mm}$$

η = 0.5 derates the MG995 to half stall (it overheats at stall). The tendon is shared by MCP, PIP and DIP, so the three joints share one tension and their excursions add up against one stroke.

3.2 The lever, from virtual work

The cleanest definition of the tendon's moment arm: if the joint closes by dθ, the tendon path shortens by dl, and the work T·dl equals the joint's work M·dθ. So

$$r(\theta) = -\frac{dl}{d\theta}, \qquad M_{tendon} = T\,r(\theta)$$

This works for any joint (pin, bending, rolling) as long as you can write the tendon length l(θ). It also says that lever and stroke are the same thing: a longer lever costs proportionally more tendon travel.

Pin joint (Type A; also used for fixed-centre Type B). Guide at (−a, h) on the proximal link, anchor at (x, h) on the distal link rotated by θ, with b = (x cos θ − h sin θ, x sin θ + h cos θ):

$$l = \|\mathbf{b} - (-a, h)\|, \qquad r = \frac{b_x h + b_y a}{l}$$

Bending joint (B3, B6, B1 full notch, lattice: roughly constant curvature over the flex length L). The tendon at offset d runs as the chord of the inner arc of radius L/θ − d:

$$l(\theta) = 2\left(\frac{L}{\theta} - d\right)\sin\frac{\theta}{2}, \qquad r = -\frac{dl}{d\theta}\ \text{(numerically)}$$

Rolling joint (B4). The contact point moves palmar by R sin(θ/2) as the lobes roll, eating into the lever:

$$r(\theta) \approx d - R\sin\frac{\theta}{2}$$

This is why B4's roll radius was kept small: with the as-built d = 1.93 mm and a large R the lever goes negative part-way through flexion and the tendon would start straightening the joint. R was set so r ≥ 0.3 mm at the target flexion. **Where in the code:** scripts/tendon_routing_analysis.py, scripts/tendon_pivot_optimizer.py, TYPE B/round2/predict_round2.py

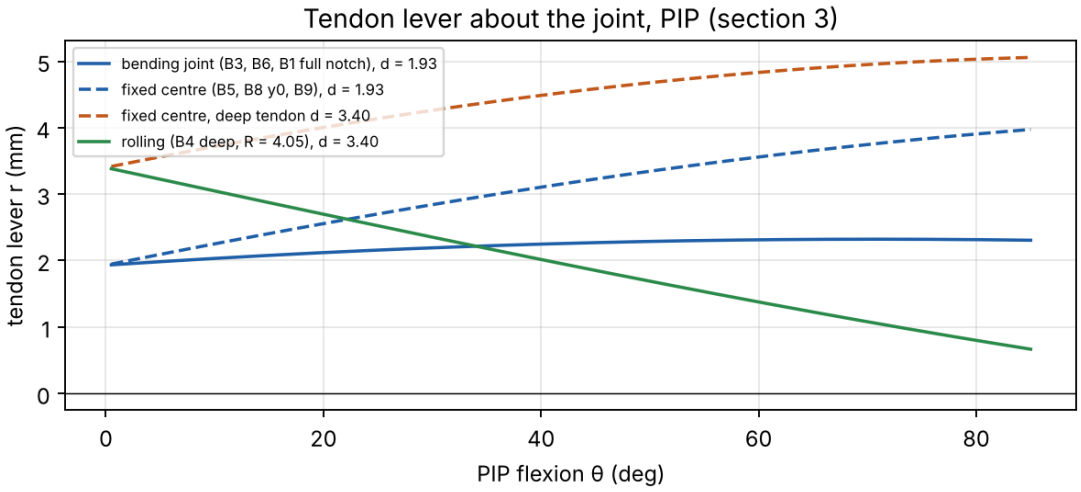


Figure 2. Lever against flexion for the joint models at the PIP (flex zone 7.55 mm), up to 85° (the collars meet at about 76°). A fixed centre gains lever as it closes; rolling loses it.

3.3 Tension to hold a load

On the single-joint rig, a weight W hangs from a hook L_{hook} = 44.8 mm from the PIP, with the distal arm raised θ above horizontal. Moment balance about the joint, including the joint's own spring k:

$$T = \frac{W\,L_{hook}\cos\theta + k\,\theta}{r(\theta)}$$

For the whole finger, a fingertip force F normal to the distal phalanx loads joint j by the lengths of all links beyond it, projected (φ_i is the cumulative angle of link i):

$$\tau_j = F \sum_{i \geq j} L_i \cos(\phi_{tip} - \phi_i)$$

Worked example: why the deep tendon offset matters
At 0° the lever equals the offset d. Tension per newton on the hook: 44.8 / 1.93 = **23.2 N/N** as built; 44.8 / 3.40 = **13.2 N/N** with the deep tendon.
So the same 1 N at the fingertip squeezes the joint with 23 N (as built) or 13 N (deep). Halving the squeeze for the same grip is the whole case for the deep-offset phalanges.
B5 t2.0 deep holds 30.2 N of tension at 0° (FEA) → 30.2 / 13.2 = **2.29 N** on the hook at 12 MPa.

3.4 Choosing tendon guide positions (Type A)

For every combination of guide station a, anchor station x and offset h, the margin is the worst ratio of available to required torque over the grasp poses:

$$\text{margin}(a, x, h) = \min_{\theta \in \text{grasp}} \frac{T_{avail}\, r(\theta; a, x, h)}{k\,\theta + \tau_{load}(\theta)}, \qquad \text{excursion} = l(0) - l(\theta_{max})$$

Keep only the Pareto front (more margin needs more excursion), then pick one point per joint to maximise the weakest joint's margin with the summed excursion ≤ 31.4 mm. Designs with lever < 0.5 mm anywhere, or guides closer than 3.5 mm, are rejected.

4. Compression capacity: hand estimates

The tendon that bends a pinless joint also squeezes it along its length. In Type A the pin takes that load. In Type B the TPU does, and it fails in one of three ways, each with its own hand formula. These were the predictions; the round-2 FEA (section 5.6) then checked them.

4.1 Thin leaves and strips: Euler buckling

A leaf held square at both ends (collars) buckles at

$$P_{cr} = \frac{4\pi^2 EI}{L^2}, \qquad I = \frac{wt^3}{12}$$

For n leaves in parallel multiply by n. A B5 crossed strip of length L_s at ±α/2 to the axis carries the axial load at an angle, adding a cos(α/2) factor; B5 has 4 lanes. B1's notch is two webs of notch length L_n.

Worked example: B3_t200_w30 and B5_t200_a45 at the PIP, E = 12 MPa

B3: n = 2, t = 2.0, w = 3.0, L = 7.55 mm → I = 3.0×2.0³/12 = 2.00 mm⁴; P = 2×4π²×12×2.00/7.55² = **33.2 N**.

B5: 4 lanes, t = 2.0, w = 1.95, L_s = 8.172 mm, α = 45° → I = 1.30 mm⁴, cos(α/2) = 0.924; P = 4×0.924×4π²×12×1.30/8.172² = **34.1 N**.

FEA: B3 buckled sideways at **10.0 N**, a third of Euler. B5 deep reached **30.2 N** (strain-limited), close to its 34.1 N estimate.

Lesson: Euler assumes perfectly straight, centrally loaded leaves. The tendon pushes off-centre and the 1 % side push finds the sideways mode, so a plain leaf buckles far earlier. A crossed strip is braced against that mode and behaves much closer to the formula.

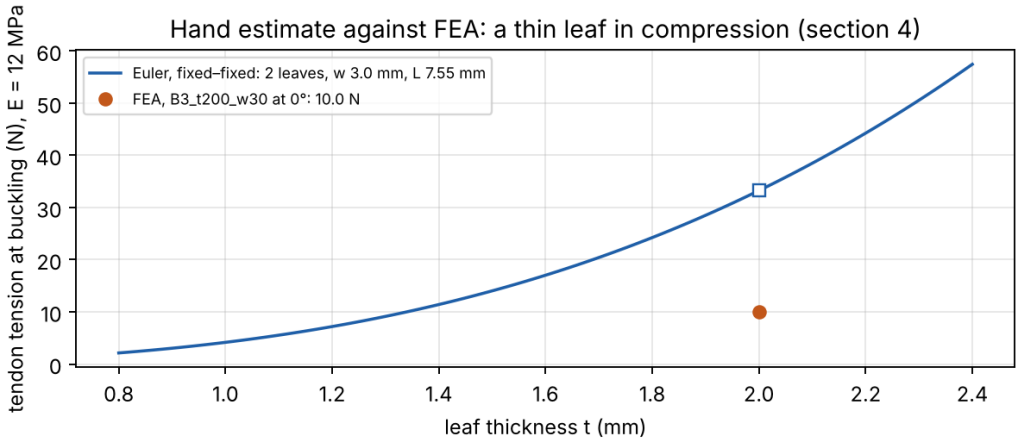


Figure 3. Euler's formula against the FEA for the B3 dual leaf. Treat hand buckling loads for single leaves as upper bounds.

4.2 Stocky struts and knuckles: 20 % strain over an area

$$T_{cap} = 0.2\,EA, \qquad A_{strut} = \text{neck area}, \qquad A_{knuckle} = 2r\,w_{bearing}$$

Worked example: B9_n150 and B8_y00_r150

B9: neck area 5.85 mm² → 0.2×12×5.85 = **14.0 N**. FEA at 0°: 12.7 N. Good agreement, but at 60° bending alone already strains the neck to 37 %.

B8: knuckle r = 1.5 mm, bearing width 3.9 mm → 0.2×12×2×1.5×3.9 = **28.1 N**. FEA reached ≥ 5.8 N (a lower bound; the contact solution stopped first).

4.3 Rolling lobes: Hertz line contact

Two TPU cylinders of radius R pressed together over width w. The combined radius is R' = R/2 and the combined modulus for two identical materials E* = E / (2(1 − ν²)). Hertz gives the peak pressure p_0 for a load P; set p_0 = 0.2E (about 20 % local strain) and solve for P:

$$p_0 = \sqrt{\frac{PE^*}{\pi w R'}} \quad \Rightarrow \quad T_{cap} = \frac{\pi w R' p_0^2}{E^*}$$

Worked example: B4_t100_a60_deep

R = 4.047 mm → R' = 2.023 mm; w = 3.9 mm; ν = 0.45 → E* = 12/(2×(1−0.45²)) = 7.52 MPa; p_0 = 2.4 MPa.

T_{cap} = π×3.9×2.023×2.4²/7.52 = **19.0 N**. FEA reached ≥ 11.1 N at 12 % strain before the contact solution stopped.

4.4 From capacity to fingertip, and the collar hard stop

$$W_{hook,\,max} = \frac{T_{cap}}{\max\limits_{\theta}\,(L_{hook}\cos\theta / r(\theta))}, \qquad \theta_{collar} = 2\arctan\frac{F_{flex}/2}{h_c + y_{pivot}}$$

For the PIP the collars (half-height h_c = 5.7 mm, flex zone F = 7.55 mm) meet at 2·atan(3.775/5.7) = 67° about a central pivot. The round-1 FEA found about 76° (conservative estimate). This, not the flexure, sets most Type B ranges; palmar chamfers on the collars would lift it.

5. The finite-element method as used here

5.1 Mesh and solver

- Solver: CalculiX (versions 2.21 and 2.23 agreed to 5 significant figures on B3_t200_w30), direct solver SPOOLES, geometric nonlinearity on (NLGEOM), Newton–Raphson with automatic increments.
- Elements: 10-node quadratic tetrahedra (C3D10) from gmsh. Rule: at least 2 elements through the thinnest feature, element size h = min(t, w)/2, floor 0.2 mm. Round-2 joints use a graded mesh (fine in the thin spring lane, 0.5 mm in bulky bearings, 0.8 mm in collars) to keep memory under 8 GB.

5.2 Boundary conditions: the part that decides whether the answer is right

- Proximal collar: every node fixed. Distal collar: **every** surface inside the attachment tied to one rigid reference node. (Tying only one face made A3 63× too soft.)
- **Type A** (spring beside a pin): the reference node sits on the pin axis, its translations are fixed and it is rotated. (Leaving translations free made A3 55× too soft, with no warning.) The axis position must be given explicitly.
- **Type B** (the TPU is the joint): rotation prescribed, translations **free**. That is a pure moment under displacement control, and the joint finds its own centre of rotation, which is reported as axis drift.
- Joint angle is applied in increments up to the target (A8 to 100°, Type B PIP to 105°); each increment records the reaction moment, so the output is a torque–angle curve.

5.3 What is extracted from a torque–angle curve

$$k_0 = \text{slope of a line fit of } M(\phi) \text{ for } \phi \le 3^\circ, \qquad U_{0\to 90} = \int_0^{\pi/2} M\, d\phi, \qquad \text{linearity} = \frac{M_{90}}{k_0\, \pi/2}$$

Worked example: A8 with collars, sagitta 2.75 (E = 19.3 MPa)

k_0 = 1.075 N·mm/rad; M_{90} = 2.809 N·mm; linearity = 2.809 / (1.075 × 1.571) = **1.66**. Above 1 means the spring stiffens as it closes, which is useful because the finger needs the most return help when most flexed.

Without collars (rigid ends) the same strip gave 3.24 N·mm: flexible collars cost 13 %.

5.4 Off-axis stiffness (round 1)

Three small linear load cases on the same mesh give lateral, torsional and axial stiffness, which convert to what a user would notice:

$$\delta_{side} = \frac{1\,\text{N} \times L_{tip}}{k_{lat}}, \qquad \phi_{twist} = \frac{1\,\text{N} \times 5\,\text{mm}}{k_{tor}}, \qquad \Delta_{axial} = \frac{T_{end}}{k_{ax}}$$

5.5 Contact

Where two surfaces may touch (A8 strip on the bone, B4 lobes, B8 knuckle in cup) a surface-to-surface penalty contact is used, with penalty stiffness ∝ E so the E-scaling rule still holds:

$$k_{penalty} = 10\,\frac{E}{h}$$

h is the local element size. The bone surface is sampled and searched with a KD-tree to find the A8 clearance (smallest distance from the strip's underside to the phalanx), which set the sagitta trade-off: 2.75 mm keeps 0.39 mm clearance; 3.00 mm keeps 0.60 mm.

5.6 Round-2 tendon-load protocol and how a capacity is read

- **Step 1, bend:** rotate the distal body to the pose (0°, 30° or 60°), translations free.
- **Step 2, squeeze:** hold the rotation and ramp a force T (up to 150 N) along the tendon's line, plus a 1 % sideways push so a symmetric joint has a buckling mode to find instead of sitting on a perfect bifurcation.
- Bearings (B4, B8) are run **guided**: the distal body is driven along the tendon line under displacement control (a free load let their thin return leaves buckle at about 0.3 N before the bearing seated). Their bend follows the bearing's own kinematics: rotation about the knuckle centre for B8; for B4 the distal lobe centre orbits the proximal one by θ/2 while the body turns by θ.

The capacity is the first of three events:

$$T_{cap} = \min\left\{ T_{limit\ point},\quad T\,|_{k_t < 0.1\,k_{t,0}},\quad T\,|_{\varepsilon_{max} = 20\%} \right\},\qquad k_t = \frac{\Delta T}{\Delta u_{along}}$$

A limit point is where the solver can find no equilibrium at a higher load (buckling or collapse). k_t is the axial tangent stiffness; $k_{t,0}$ is the median over the first 5 % of increments. If bending to the pose already exceeds 20 %, the capacity is 0 (the joint cannot reach that angle). A guided run that stops before any of these is reported as a **lower bound** (≥). **Where in the code:** TYPE B/round2/fea_round2.py

5.7 Mesh convergence

Run the same case at three element sizes and compare the quantity you will use:

$$\text{spread} = \frac{\max_h X_h - \min_h X_h}{X_{h_{ref}}}$$

Worked example: A8 strip, torque at 90°

h = 0.5 / 0.4 / 0.3 mm → 3.228 / 3.236 / 3.221 N·mm → spread = (3.236 − 3.221) / 3.236 = **0.45 %**. Converged; 0.4 mm is fine.

Peak strain moves more (15.6 / 16.5 / 16.1 %) because it is a local maximum, not an integral. Judge strain against the limit with about ±1 point of mesh uncertainty.

6. Turning results into a choice

6.1 Gates: pass or fail before any ranking

Gate	Formula	Threshold	Why
Strain	peak principal engineering strain over the usable range	≤ 20 %	fatigue life of TPU (placeholder)
Return	margin = M(20°) / $M_{gravity}$	≥ 1	no extensor: the spring must lift the distal finger's own weight
Range	first of collision, 20 % strain, self-contact, solver stop	near the joint's target	the joint must close
Tendon squeeze (Type B)	T_{cap} vs T needed for the grip	T_{cap} above it	otherwise it buckles before gripping
Fit (Type A)	clearance to the bone, axis distance	> 0.25 mm; ≥ 3.5 mm	it must not rub or cut through the joint

$$M_{gravity} = \sum_i m_i\, g\, x_i \qquad (\text{distal chain horizontal}:\ \text{MCP } 3.68,\ \text{PIP } 0.80,\ \text{DIP } 0.15\ \text{N mm})$$

6.2 Ranking within a family (round 1)

Each criterion is normalised so the family's best = 1 and worst = 0, with a minimum spread so tiny differences count as ties (3° of range, 2 points of strain, 0.3 N, 1°, 1°, 0.2 mm, 0.03 mm), then combined with weights:

$$s_c(v) = \max\!\left(0,\ 1 - \frac{|x_{best} - x_v|}{\max(|x_{best} - x_{worst}|,\ \text{floor}_c)}\right), \qquad S(v) = \sum_c w_c\, s_c(v)$$

Balanced weights: range 0.30, strain 0.25, tendon force 0.15, sideways 0.10, drift 0.10, twist 0.05, compression 0.05. The winner was re-checked under three other weightings (range-first, durability-first, equal). A choice that holds under all four does not depend on the weights. **Where in the code:** TYPE B/analysis/analyse_results.py

6.3 How the print list followed from the numbers

- **A8 with collars**: the strongest Type A spring inside 20 % strain on the real PIP; torque converged; with collars 1.75 N·mm at 90° at 12 MPa. Sagitta 3.00 kept as well: 0.60 mm clearance and 19.9 % junction strain for 2.5 % less torque.
- **A6**: passes all four axis positions by the elastica; R28/1.4 has 3.2 points of strain margin, R32/1.8 sits at 19.8 %, so R28 is printed first.
- **B5 t2.0 deep**: the only Type B joint whose resolved capacity (30.2 N at 0°, 10.3 N at 60°) clears the grip loads by a useful margin.
- **B4, B8**: the bearing route is the only way past the flexure-buckling limit; the FEA could give only lower bounds, so the print is the test.
- **Set aside**: B1, B9 (strain-limited just bending to 60°), B6 and the lattice (failed earlier gates), B3 (buckles at 10 N).

7. Geometry operations used along the way

7.1 Fitting one frame onto another (Kabsch)

To place the A3 collars on the A8 strip, the same strip vertices were found in both frames and the best rotation R and translation t computed:

$$H = \sum_i (p_i - \bar{p})(q_i - \bar{q})^T = U\Sigma V^T, \qquad R = V\,\text{diag}(1, 1, \det VU^T)\,U^T, \qquad t = \bar{q} - R\bar{p}$$

Residual 4.6×10^{-12} mm: an exact fit. **Where in the code:** scripts/a8_collared_geometry.py

7.2 Instantaneous centre of rotation

For a body point P moving with velocity v at unit rotation rate, the centre is c = P + ($-v_y$, v_x). For a constant-curvature joint P(θ) = (L/θ · sin θ, L/θ · (1 − cos θ)), differentiated numerically. Its drift from mid flex zone is a ranking criterion because a wandering centre changes the tendon lever.

8. Recreating the work yourself

Order of operations for a new joint idea, from minutes to hours:

Step	What	Tool / command	Time
1	Strain-size the spring: t from $\varepsilon = t\Delta\kappa/2 \le 0.2$, then w for torque	by hand (section 2.1)	minutes
2	Tendon lever and tension per newton at 0°, 30°, 60°	python "TYPE B/round2/predict_round2.py"	seconds
3	Compression capacity by hand (Euler / strain / Hertz)	same script	seconds
4	Large-deflection beam if it bends a lot (elastica)	python scripts/a6_rom_check.py	minutes
5	Bending FEA: torque–angle, strain, range, return margin	python "TYPE B/analysis/fea_typeb.py" <variant> PIP	2–80 min
6	Tendon-load FEA at 0/30/60°	python "TYPE B/round2/fea_round2.py" <variant> --angle 60	0.5–2 h
7	Type A spring on the real bone, with contact	python scripts/a8_meander_study.py --groups ...	~2 h per case
8	Mesh check on the chosen design (3 sizes)	--printable-element-size 0.5 / 0.4 / 0.3	hours
9	Put everything on one E and rank	python scripts/sunlu_rescale.py; python "TYPE B/analysis/analyse_results.py" PIP	seconds
10	Rebuild the reports	python scripts/build_print_report.py; python scripts/build_math_report.py	a minute

8.1 Mistakes that produce plausible but wrong numbers (all happened here)

- Joint axis at the bounding-box centre instead of the real joint: A3 stiffness halved. Always give the axis explicitly.
- Tying one face of a collar instead of all of it: 63× too soft. Leaving a pinned joint's translations free: 55× too soft.
- Measuring a section with a cut that is not perpendicular to the member: 5 % too thick at 17°, so 16 % too stiff.
- Clamping a beam model 0.5 mm shorter than the FEA: most of a reported 24 % disagreement.
- Trusting Euler for a single leaf: 3× too optimistic (section 4.1).
- Reporting a solver stop as 'capacity 0': it means 'not resolved', not 'no strength'. Say which.
- Comparing torques from studies at different E without rescaling (section 1.2).

8.2 When the specimens are tested

- Tensile 1BA bars: E = slope of stress against strain from 0.5 % to 2 %, mean of at least 3 per thickness. Set E_SUNLU_MPA.
- Ø10 × 10 compression cylinder: E in compression from 2–10 % strain. The B4/B8 bearings depend on this one.
- If the stress–strain curve bends noticeably before 20 %, fit Neo-Hookean (or two-term Mooney–Rivlin) with HyperelasticModel.fit_uniaxial and re-run the few chosen designs. Rankings will not change; absolute numbers will.
- Cycle a strip to find the real strain limit, and replace the 20 % everywhere it appears (STRAIN_LIMIT in the scripts).

Appendix: symbols

Symbol	Meaning	Unit
E, ν, μ_0, C_{10}	Young's modulus, Poisson's ratio, initial shear modulus, Neo-Hookean constant	MPa, –
t, w, L, I	thickness (through the bend), width, free length, second moment $w\,t^3/12$	mm, mm⁴
κ, κ_0, ε	curvature, initial curvature, engineering strain	1/mm, –
θ, φ	joint flexion angle	rad (deg in tables)
M, τ, k, k_0	moment / torque about the joint; joint stiffness; initial stiffness	N·mm, N·mm/rad
T, r, d	tendon tension, tendon lever about the joint, tendon offset from the neutral axis	N, mm
L_{hook}, W	rig hook distance (44.8 mm), hook load	mm, N
P_{cr}, T_{cap}	buckling load, compression capacity in tendon tension	N

Symbol	Meaning	Unit
R, R', E*	roll radius, combined radius R/2, combined contact modulus	mm, MPa
h	element size	mm